



EE6620 計算攝影學

Computational Photography

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加簽方式

- 上課人數目標: ≤ 40 人
- 加簽進行方式(線上加簽):
 1. 學生於2/25 9PM前，到選課系統上申請加簽
 2. 老師於2/26 9AM前，決定核可加簽對象
 - 若欲加簽人數超過預期，會以加簽理由來判斷

Imaging revolution is continuing...

- Traditional photography
 - Optical lenses focus rays onto sensor
 - Chemistry records final image
- Digital photography
 - Optical lenses focus rays onto sensor
 - Opto-electronic sensor records final image

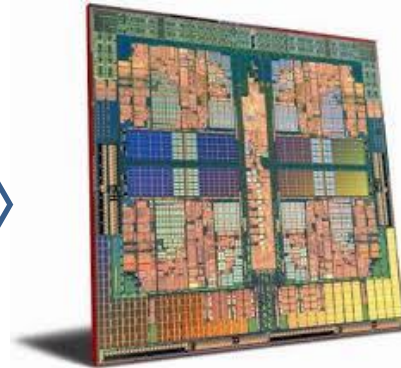


Imaging revolution is continuing...

- Computational photography
 - Computation between recorded digital samples and final image



Specialized imaging
(e.g. burst imaging)



(Lots of) Computation



Final image
(e.g. super-resolution, HDR)



Outline

- **Course facts**
- Relationship to CV and ML
- Quick view of course content



天時、地利、人和

- Class Hours: W789
- Lecture Room: 台達 211
- Instructor: 黃朝宗
- TA: 陳佳萱、林庭仔



課程目標

- 介紹如何透過計算來增強拍照的影像品質，並實際藉由程式實現來驗證可行性。
- 介紹如何將實際影像應用以數學式建立模型，並透過程式來實現處理過程。
- 除了讓學生了解先進的影像處理演算法外，亦透過Python學習如何使用影像處理函式庫，如影像最佳化的ProximalL，以及卷積神經網路框架PyTorch。

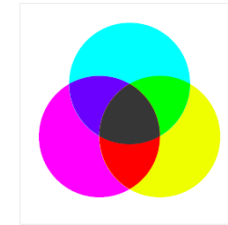
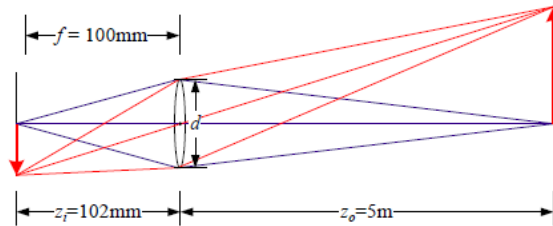


授課方式

- 本課程依計算攝影學不同主題分為四單元：相機成像、以最佳化為基底之影像處理、以卷積神經網路為基底之影像處理、其它特選主題(2026: Diffusion Model)。
- 除了講堂上以投影片授課以外，有三次Python程式作業(對應前三單元)，與期末專題，以透過解決實際影像問題來體驗計算攝影學的威力。

教學內容

- Part I: Image formation
 - Two weeks, HW1 (Python)



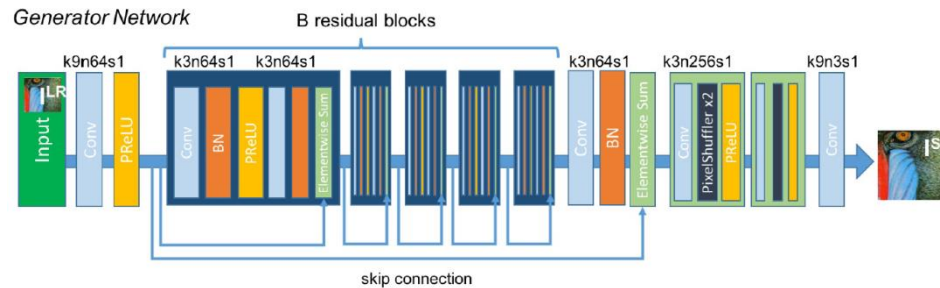
- Part II: Optimization-based processing
 - Three weeks, HW2 (Python+ProxImaL)



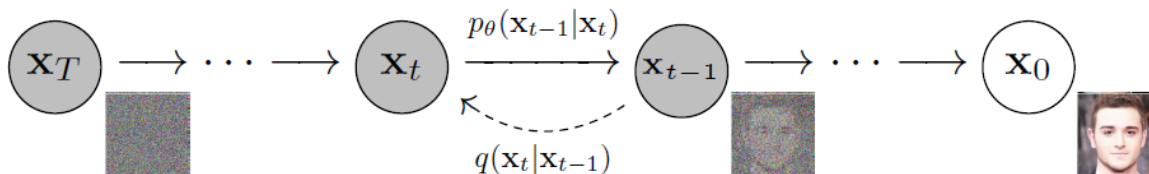
$$\min_{\mathbf{x}} \|\mathbf{y} - \Phi\mathbf{x}\|_2^2 + \lambda\mathcal{R}(\mathbf{x})$$

教學內容

- Part III: Convolutional neural networks
 - Five weeks, HW3 (ProxImaL+PyTorch)



- Part IV: Selected topics (diffusion model)
 - Four weeks





期末專題

- May 6: Project announcement
- May 20: Interim presentation
- Jun 10: Final presentation, final report
- 學生有責任說明Project的內容有哪裡是由自己發想與撰寫



課程行事曆

L01: Syllabus and introduction
L02: Photometric image formation and digital camera
L03: Color space
L04: High dynamic range imaging
L05: Image denoising
L06: Image deblurring
L07: Optimization-based image processing
L08: Super-resolution
L09: Deep convolutional network
L10: CNN-based image restoration
L11: CNN for image processing
L12: Generative adversarial network
L13: Towards real-scene image restoration
L14: Diffusion model

Week	W789	Lecture (台達211)	HW out	HW due
1	2/25	L01; L02		
2	3/4	L03; L04		
3	3/11	L05; Python	1	
4	3/18	L06		
5	3/25	L07; L08		
6	4/1	L09	2	1
7	4/8	L09		
8	4/15	L10; L11		
9	4/22	L11; L12; PyTorch	3	2
10	4/29	L12; L13		
11	5/6	L14		
		Project Announcement		
12	5/13	L14		3
13	5/20	L14		
		Interim Presentation		
14	5/27	L14		
15	6/3	No class		
16	6/10	Final Presentation		

16週因應方案: 遇國定假日或特殊狀況(例如老師出國開會), Lecture會改成預錄視訊, HW照出, 若有需要會在前一週先進行說明。



成績考核

- Homework

- 這些是個人型作業，禁止未經許可的合作(Unauthorized Collaboration)
- 學生有責任證明作業的程式是由自己發想與撰寫
- 如果有被判定為共謀(Collusion)的狀況，等同學術作弊，那麼長的很像的作業們，成績會除於共謀人數
 - 例如五個人的作業長的很像，而且都滿分，那每人的成績會是 $100/5=20$
- 可以遲交，但每超過一天，分數就打九折一次
- 作業繳交時間以eeclass紀錄為準，且以該次上傳內容評分
 - 若上傳錯內容，就得重新上傳，並以新的繳交時間為準

- 評比比重

- Homework (60%)
 - HW1/2/3: 20%/20%/20%
- Term Project (40%)



課程網頁

- eeclass website
 - <https://eeclass.nthu.edu.tw/course/30976>
 - 作業與期末專題皆繳交於此



Outline

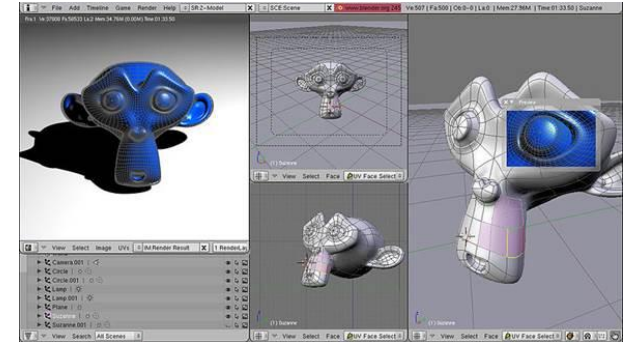
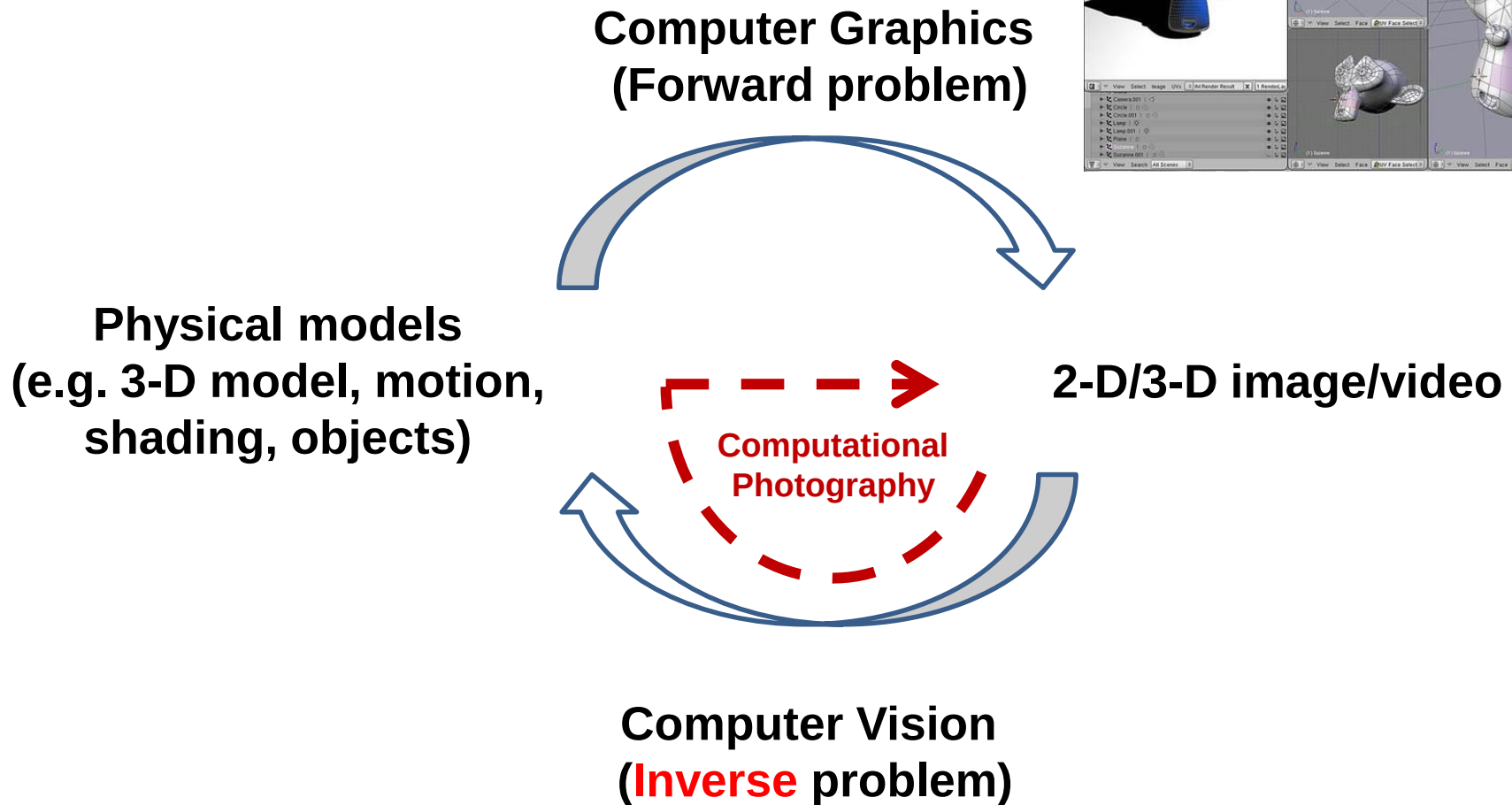
- Course facts
- **Relationship to CV and ML**
- Quick view of course content

Computer vision (CV)

- Vision
 - How we humans understand and describe the world that we see in the projected images on the retina
- Computer Vision
 - Understand and describe this world through captured images/videos with computational techniques

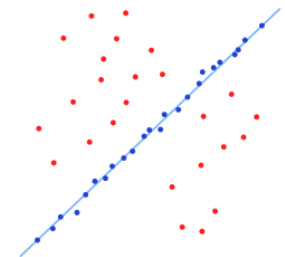
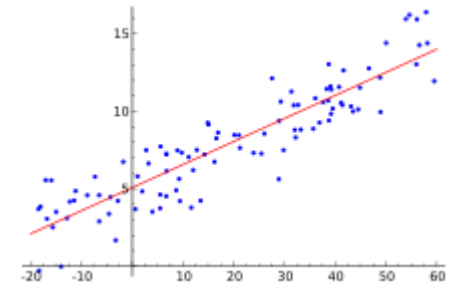
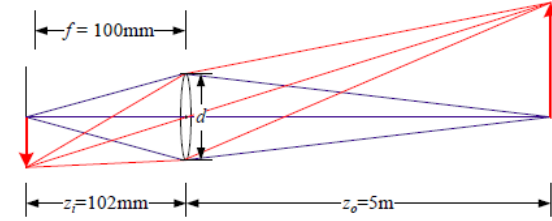


Computer graphics vs. Computer vision



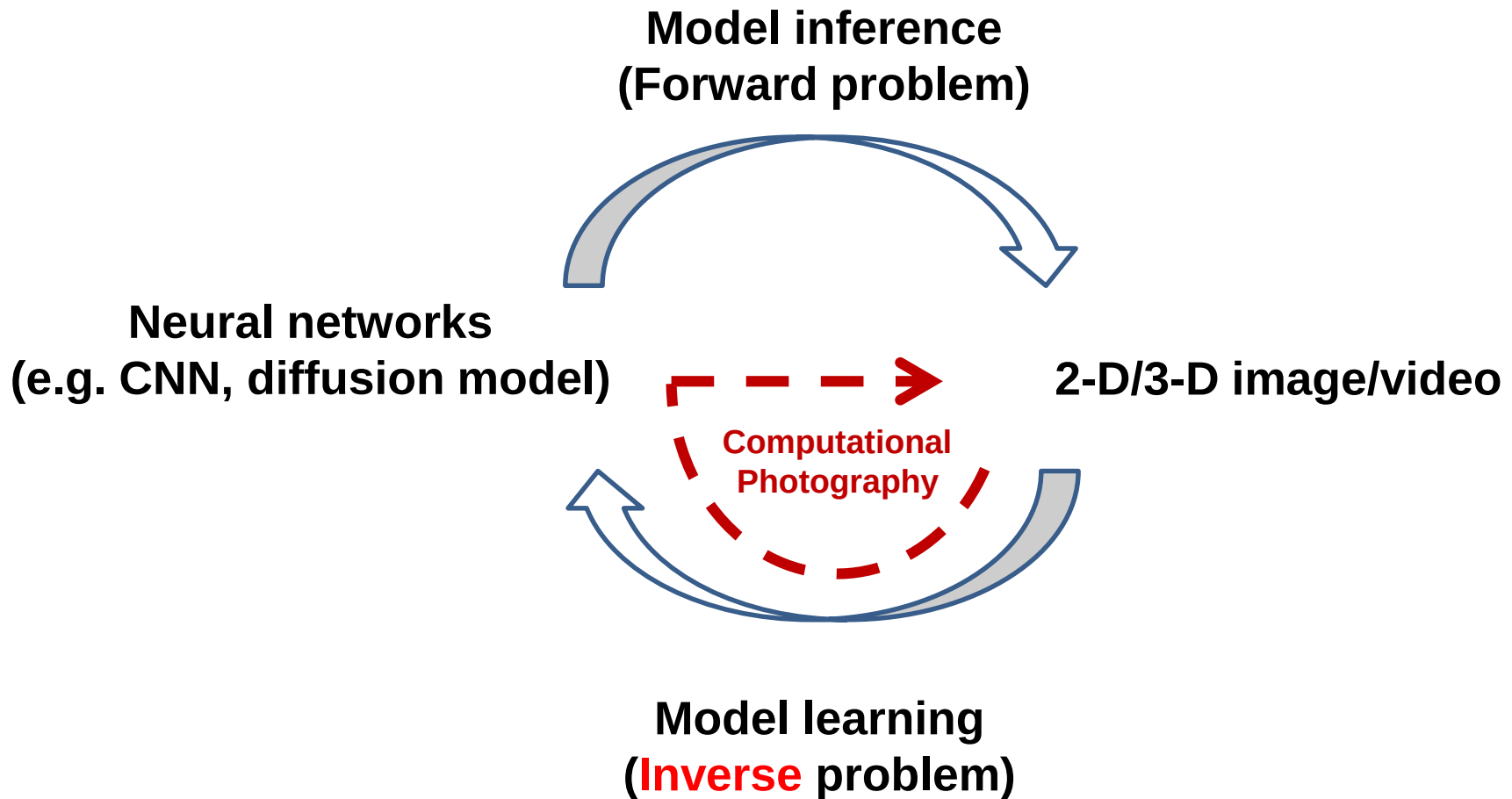
Three high-level approaches for CV

- Scientific
 - Physical models in formulations
 - Testable ideas (abstraction)
- Statistical
 - Probabilistic models for inference
 - Model-observation bridge (reality)
- Engineering
 - Robust implementation in practice
 - Robustness-efficiency tradeoff





Machine-learning based processing





Three high-level approaches for ML

- Data-driven
 - Parameterized neural network models
 - Non-linear formulations (abstraction)
- Statistical
 - Probabilistic models for learning and inference
 - Model-observation bridge (reality)
- Engineering
 - Robust implementation in practice
 - Robustness-efficiency tradeoff



**“All models are wrong,
but some are useful.”**

by George E. P. Box

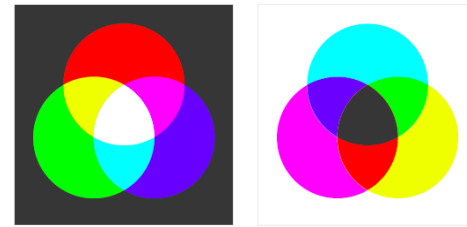
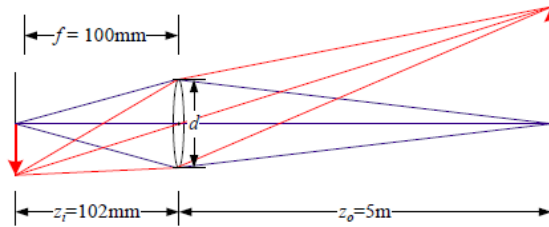


Outline

- Course facts
- Relationship to CV and ML
- **Quick view of course content**

Part I: Image formation

- Photometric formulation, digital camera, color space



- HDR



Part II: Optimization-based

- Denoise

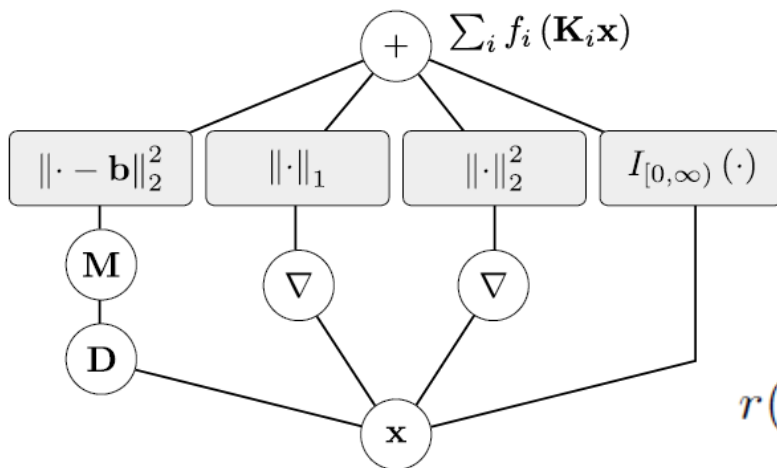


- Deblur



Part II: Optimization-based

- Super-resolution
- Optimization framework

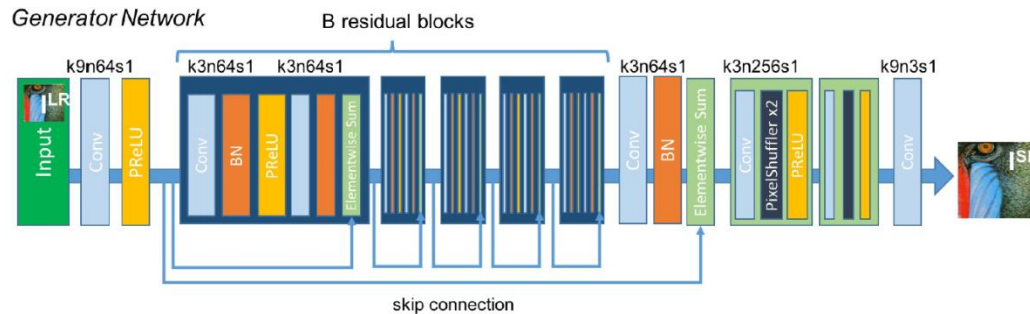


$$\mathbf{x}_{\text{opt}} = \underset{\mathbf{x}}{\operatorname{argmin}} \quad \|\mathbf{MD}\mathbf{x} - \mathbf{b}\|_2^2 + r(\mathbf{x})$$

$$r(\mathbf{x}) = \mu \|\nabla \mathbf{x}\|_1 + (1 - \mu) \|\nabla \mathbf{x}\|_2^2 + I_{[0, \infty)}(\mathbf{x})$$

Part III: Convolutional Networks

- Introduction to CNN and GAN
- CNN-based super-resolution

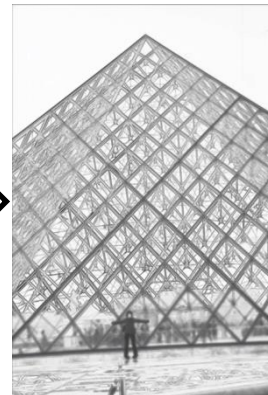


- Other image applications

Monet $\leftrightarrow$ Photos



Zebras $\leftrightarrow$ Horses

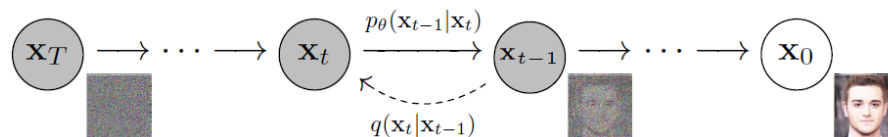


Part IV: Selected topics

- Diffusion model: application

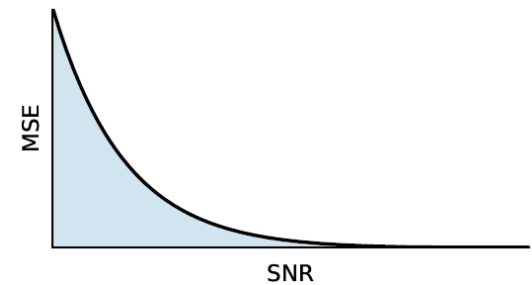
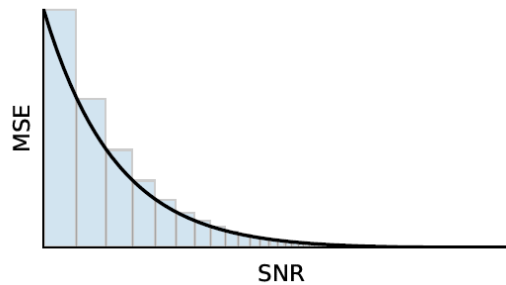
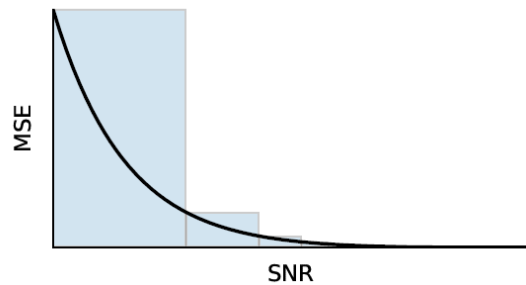
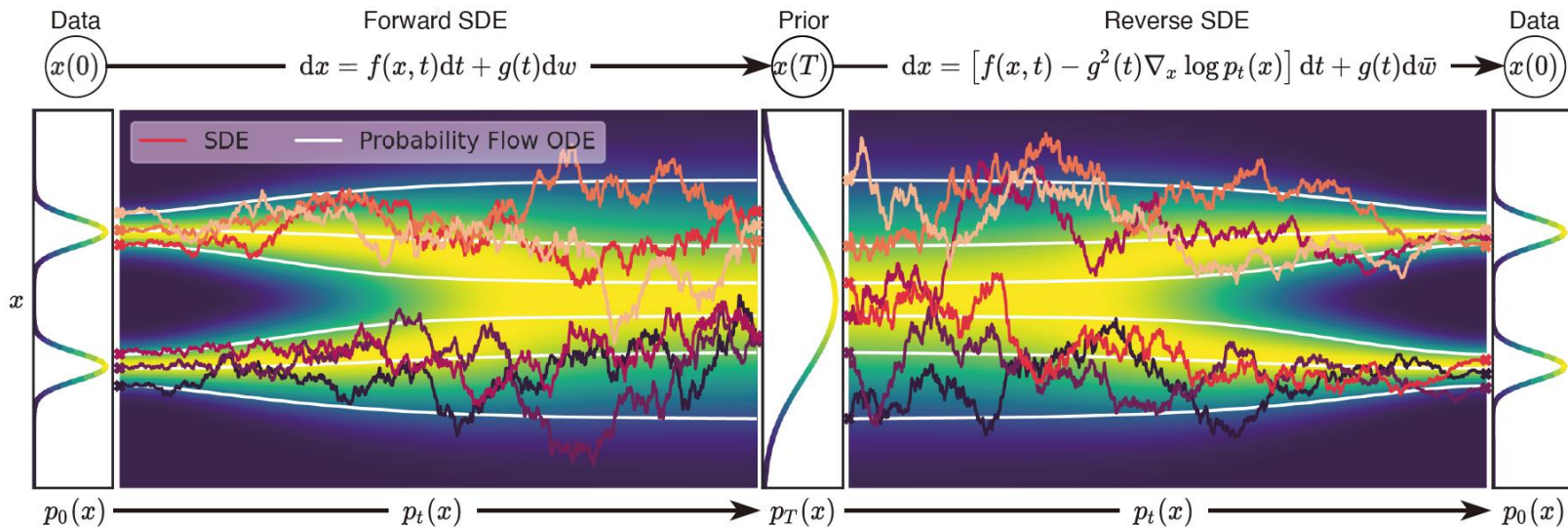


A cute corgi lives in a house made out of sushi.



Part IV: Selected topics

- Diffusion model: theory



Part IV: Selected topics

- Diffusion model: implementation

